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MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL

Paper Code M-401

MATHEMATICS – III

Time Allotted : 3 Hours | *Full Marks* : 70

The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable.

GROUP – A

(Multiple Choice Type Questions)

1. Choose the correct alternatives for any *ten* of the following : $10 \times 1 = 10$

i) A fair die is thrown 180 times. The expected number of sixes is

a) 10 b) 20

c) 30 d) 40.

ii) Let G be a Group and $x \in G$ be such that $o(x) = 5$. Then

a) $o(x^{15}) = 4$ b) $o(x^{10}) = 6$

c) $o(x^{23}) = 5$ d) $o(x^{20}) = 3$.

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iii) The set {[2], [4], [6], [8]} is a group under multiplication modulo 10. The identity element of the group is

a) [2] b) [4]

c) [6] d) [8].

iv) let G be a finite group of even order. Then the number of elements of order 2 in G is

a) 2 b) 4

c) even d) odd.

v) The probability that *A* passes the exam is 0·9 and *B* passes the exam. is 0·8. The probability that at least one of them passes is

a) 0·98 b) 0·97

c) 0·9 d) 0·72.

vi) Let X be a random variable and λ be a real number. Then $\frac{Var(\lambda X)}{Var(X)}$ is

a) λ b) λ^2

c) $\frac{1}{\lambda}$ d) $\frac{1}{\lambda^2}$.

vii) If $E(T) = 5\theta + 6$ then the unbiased estimator of θ is

a) $\frac{1}{5}T - \frac{6}{5}$ b) $T - \frac{6}{5}$

c) $\frac{1}{5}T$ d) $T - 2$.

viii) If G is a non-planar graph then the possible number of vertices of G is

a) 2 b) 3

c) 4 d) 6.

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ix) The chromatic number of a graph containing an odd circuit is

a) 3

b) 2

c) greater than or equal to 3

d) greater than or equal to 2.

x) The generators of the cyclic group (Z, +) are

a) 1, - 1 b) 0, 1

c) 0, - 1 d) 2, - 2.

xi) In the group S_3 of all permutations on {1, 2, 3}; the inverse of $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}$ is

a) $\begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix}$ b) $\begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix}$

c) $\begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}$ d) $\begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix}$.

xii) The number of unit elements of the ring (Z, +, .) is

a) 2 b) 3

c) 1 d) infinite.

xiii) For a random variable X with mean 0, the value of $P(-4\sigma < X < 4\sigma)$ will be at least

a) $\frac{1}{16}$ b) $\frac{1}{4}$

c) $\frac{15}{16}$ d) $\frac{13}{16}$.

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xiv) Corresponding to a pendant edge in G we get

a) two parallel edges in its dual

b) a loop in its dual

c) an isolated vertex in its dual

d) another pendant edge in its dual.

GROUP – B

(Short Answer Type Questions)

Answer any *three* of the following $3 \times 5 = 15$

2. Let $(G, *)$ be a group and $a, b \in G$. Suppose that $a^2 = e$ and $a * b^4 * a = b^7$. Prove that $b^{33} = e$. 5

3. If $X_1, X_2, X_3, ..., X_n$ constitute a random sample of size n from an infinite population with mean μ and variance σ^2, then prove that $E(\overline{X}) = \mu$ and $Var(\overline{X}) = \frac{\sigma^2}{n}$. 5

4. Prove that every finite integral domain is a field. 5

5. Prove that a planer graph G with n vertices, e number of edges and k number of connected components determines $f = e - n + k + 1$ number of regions. 5

6. Find out order of the element $(1\ 2\ 3) \in S_3$. Also find out inverse of $(1\ 2\ 3) \in S_3$. 3 + 2

7. A speaks truth 4 out of 5 times. A die is tossed. He reports that there is a six. What is the probability that it was actually a six ? 5

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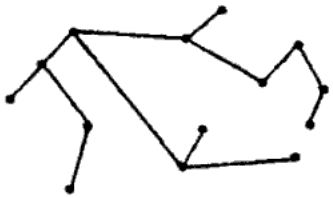
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GROUP – C

(Long Answer Type Questions)

Answer any *three* of the following. $3 \times 15 = 45$

8. a) Find out the chromatic polynomial of the following graph G :



5

b) Let H be a subgroup of a finite group G. Prove that the order of H divides the order of G. 5

c) Show that the group $(\mathbb{Z}_6, +)$ is a homomorphic image of the group $(\mathbb{Z}, +)$. 5

9. a) Prove that if G is a simple planar graph then G has at least one vertex v such that $deg(v) \leq 5$. 5

b) Prove that every group of prime order is cyclic. 5

c) Show that center of a group G, given by

$Z(G) = \{ a \in G : a * g = g * a \text{ for all } g \in G \}$

is a normal subgroup of G. 5

10. a) Experience shows that 20% of the people reserving tables at a certain restaurant never show up. If the restaurant has 50 tables and it takes 52 reservations, then find the probability that it will be able to accommodate everyone. 5

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b) The mean yield for one acre plot is 662 kilos with a standard deviation 32 kilos. Assuming normal distribution how many one acre plots in a batch of 1000 plots would you expect to have yield over 700 kilos ? (Given that $\phi(1.19) = 0.3830$) 5

c) A regular graph G determines 8 regions, degree of each vertex being 3. Find the number of vertices of G. 5

11. a) Let X be uniformly distributed over the interval $[1, 2]$ and $\overline{X} = E(X)$. Find out the value of a so that $P(X > a + \overline{X}) = \frac{1}{6}$. 5

b) A random sample of size $n = 100$ is taken from an infinite population with the mean $\mu = 75$ and the variance $\sigma^2 = 256$. Based on Chebyshev's theorem with what probability can we assert that the value we obtain for $\overline{X}$ will fall between 67 and 83 ? 5

c) If X has the standard normal variate, then find the probability density function of $Z = X^2$. 5

12. a) Let $S_1 = \{ z \in \mathbb{C} : |z| = 1 \}$ is a subgroup of $(\mathbb{C}^*, .)$ where $\mathbb{C}^* = \mathbb{C} \setminus \{0\}$. Also consider the group $(\mathbb{R}, +)$. Define a map $f : \mathbb{R} \to \mathbb{C}^*$ by $f(x) = e^{ix}$. Prove that f is a homomorphism and find out $ker(f)$. 5

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b) A random sample with observations 65, 71, 64, 71, 70, 69, 64, 63, 67, 68 is drawn from a normal population with a standard deviation $\sqrt{7.056}$. Test the hypothesis that the population mean is 69 at 1% level of significance.

[Given that $P(0 < z < 2.58) = 0.495$]. 5

c) Let $x_1, x_2, ..., x_n$ be the values of a random sample from an exponential population that is $f(x_i) = \frac{1}{\theta} e^{-\frac{x}{\theta}}$ for $x_i > 0$. Then find the maximum likelihood estimator of its parameter θ. 5

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